Chapter test

36 Human impacts on the environment

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** an impact of rapid population growth?

   A. depletion of natural resources
   B. increase in the demand for food
   C. increase in life expectancy
   D. environmental deterioration ☐

2. Which of the following are non-renewable resources?

   (1) aluminium
   (2) timber
   (3) coal

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

3. Soil erosion may be speeded up by

   (1) deforestation.
   (2) overgrazing.
   (3) growing a single crop species for a number of years.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

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★ 4. Which of the following are the harmful effects of reclamation on the environment?

(1) release of toxic heavy metals into the water
(2) destruction of habitats
(3) algal bloom

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

5. Which of the following is ***not*** a possible cause of global warming?

A. deforestation
B. acid rain
C. burning of fossil fuels
D. rearing livestock ☐

6. In industrial cities, many trees are killed due to serious air pollution. Which of the following air pollutants may be the major cause of death of these trees?

A. methane
B. carbon monoxide
C. carbon dioxide
D. sulphur dioxide ☐

★ 7. Which of the following conditions may lead to algal bloom?

(1) high concentration of nitrate in sewage
(2) high concentration of phosphate in sewage
(3) high concentration of dissolved oxygen in sewage

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

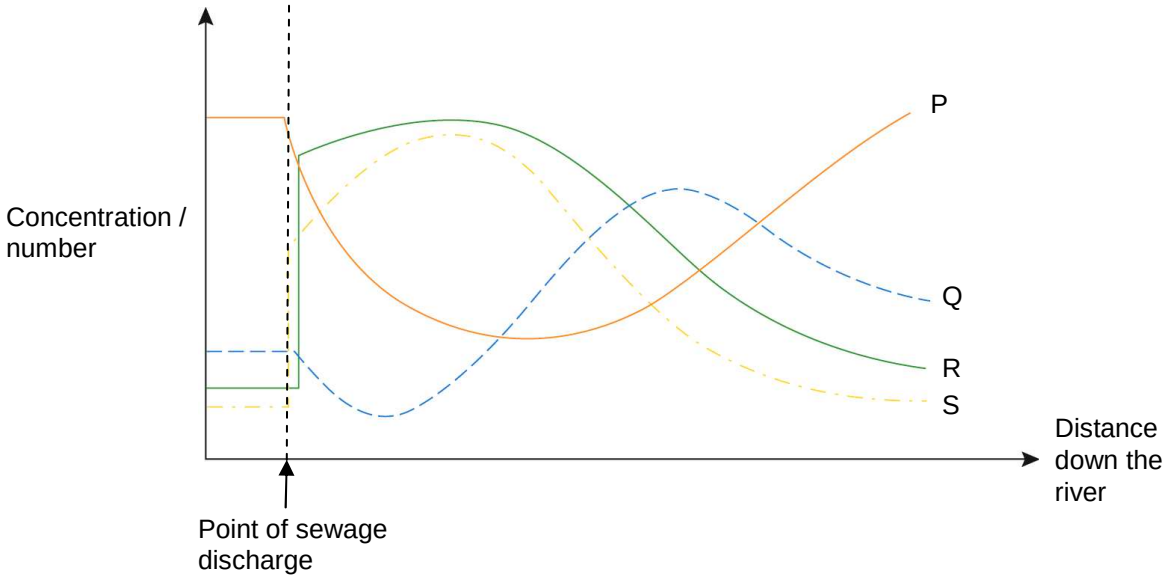
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★ 8. Algal bloom may harm aquatic organisms by

(1) producing toxins.
(2) depleting the dissolved oxygen in the water.
(3) depleting the inorganic nutrients in the water.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★★9. Untreated sewage from a farm is discharged into a river. The graph below shows the concentrations of dissolved oxygen, ammonia, nitrate and the number of bacteria down the river.



Which curves represent the concentration of dissolved oxygen and number of bacteria in the river respectively?

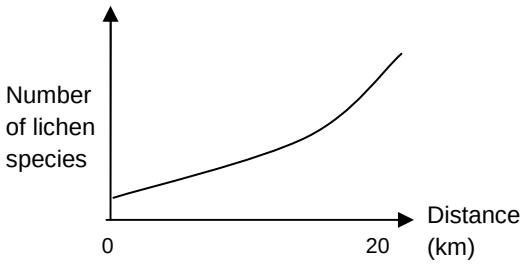
	Concentration of dissolved oxygen	*Number of bacteria*
A.	Q	P
B.	P	S
C.	R	S
D.	R	Q

☐

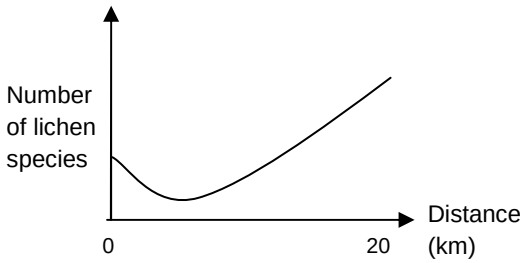
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★ 10. Lichens are sensitive to sulphur dioxide. They grow poorly in the area with high concentrations of sulphur dioxide. Which of the following graphs represents the changes in the number of lichen species from the centre of an industrial city to a rural area 20 km away?

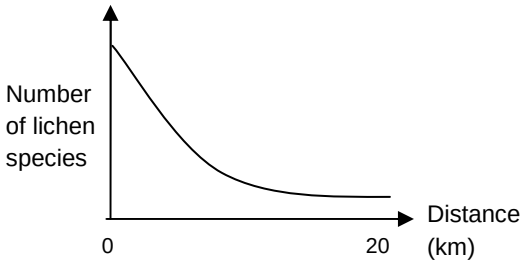
A.



B.



C.



D.



□

Question	1	2	3	4	5	6	7	8	9	10
Related section	36.1	36.2	36.4	36.6	36.6	36.6	36.6	36.6	36.4	36.6

B. Problem solving (17 marks)

11. Read the following passage and answer the questions that follow.

Use of pesticides can have unintended effects on the environment. Excessive use of pesticides can cause water pollution and decrease in biodiversity.

Besides, pesticides can also affect human health. The health effect of pesticides depends on the type of pesticide used. Some affect the nervous system while others may irritate the skin or the eyes. Some pesticides are carcinogens and some may affect the hormonal or endocrine system in the body.

From the 1950s to the 1960s, DDT became one of the most widely used pesticides. DDT is a chemical pesticide which is insoluble in water but soluble in lipid. In humans, it interferes the spread of nerve impulse and loss of control of muscle may be resulted. Therefore, the use of DDT was banned in the United States in 1972.

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(a) Suggest ***two*** possible uses of pesticides. (2 marks)

(b) With reference to the solubility of DDT, in what tissue of the animal body would DDT be accumulated? (1 mark)

(c) Explain why DDT is more harmful to aquatic animals in a lake than to aquatic animals in the ocean. (1 mark)

(d) How does DDT affect the consumers at higher trophic levels, such as birds and humans? (4 marks)

(Total: 8 marks)

★ 12. The photographs below show the effect of acid rain on trees.



Trees not affected by acid rain



Trees affected by acid rain

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(a) With reference to the above photographs, describe and explain how acid rain affects the growth of trees. (4 marks)

(b) Describe the harmful effects of acid rain on humans and other organisms. (5 marks)

(Total: 9 marks)

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C. Exam challenge (13 marks)

★ 13. Activities of reclamation increase the level of suspended solids in water.

(a) What is the purpose of conducting reclamation? (1 mark)

(b) State and explain ***two*** impacts of the reclamation work on the environment and organisms. (4 marks)

The photograph below shows a silt curtain surrounding a reclamation site in Hong Kong. It is erected from the bottom of the sea, enclosing the working site.



(c) Suggest the use of the silt curtain in the reclamation site. (2 marks)

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(d) The photograph above also shows a smoggy condition, which is caused by various air pollutants. Suggest ***two*** human health problems that may be caused by prolonged exposure to the smog. (2 marks)

(e) (i) Briefly describe how different air pollutants contribute to the formation of acid rain. (3 marks)

(ii) Acid rain is an environmental problem arising from local air pollution. But it can also cause problems to distant locations. Explain why. (1 mark)

(Total: 13 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	B	3.	D	4.	A	5.	B
6.	D	7.	A	8.	A	9.	B	10.	A

(1 mark each)

B. Problem solving (17 marks)

11. (a) To kill insect pests that feed on crops to increase crop yield. (1)
To kill mosquitoes or other disease vectors that spread infectious diseases (e.g. malaria). (1)

(b) Fatty tissues (1)

(c) The aquatic animals in the lake have a more limited choice of prey. Their chance of feeding on prey with DDT is higher. (1)
(Or other acceptable answers)

(d) When DDT is washed away from farmland into rivers, it may be taken up by aquatic organisms. (1)
DDT cannot be excreted / metabolized by aquatic organisms and thus accumulates in their bodies. (1)
DDT is transferred to the next trophic level through feeding. (1)
DDT continues to accumulate along the food chain, causing harm to the consumers when its concentration reaches high levels. (1)

(Total: 8 marks)

12. (a) Trees affected by acid rain have fewer leaves. (1)
Acid rain damages the cuticle of the leaves and makes the leaves vulnerable to microbial infection. Therefore, the trees become very weak or even die. (1)
Trees affected by acid rain have poor growth as they have fewer leaves to produce food. (1)
Acid rain dissolves minerals in the soil, allowing them to be leached away readily. (1)
Soil fertility is lowered as a result. Fewer nutrients are available for the plants. (1)
(Or other acceptable answers)

(b) Any ***five*** of the following: (max. 5)
- Breathing in acid rain particles aggravates human respiratory diseases such as bronchitis and asthma. (1)
- Acid rain causes damage to buildings and monuments because marble and various metals are corroded by acidic rain water. (1)

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- Acid rain can increase the release of aluminium ions from soil. These ions may be washed into water bodies and poison aquatic organisms. (1)
- Acid rain causes a decrease in the pH of water and harms aquatic organisms that are sensitive to pH. (1)
- Acid rain may affect the reproduction of some aquatic species. (1)
- When trees are damaged by acid rain, animals may lose their habitats and face food shortage. (1)

(Or other acceptable answers)

(Total: 9 marks)

C. Exam challenge (13 marks)

13. (a) To obtain more land for urbanization / industrialization. (1)

(b) Reclamation would alter the natural coastline. (1)
This leads to destruction of habitats and loss of biodiversity. (1)
The suspended solid particles may clog the gills of fish, hence reducing their respiratory efficiency. (1)
The water will become turbid and thus reduces light penetration and photosynthesis of seaweeds. This reduces the amount of energy entering the marine ecosystem, which affects other organisms throughout the food web. (1)
(Or other acceptable answers)

(c) The slit curtain prevents suspended solids generated in the reclamation site from spreading to adjacent water. (1)
This helps keep the water quality in adjacent water. (1)

(d) The smog irritates the eyes / nose / breathing system. (1)
The smog worsens existing heart conditions / respiratory diseases (e.g. asthma). (1)

(e) (i) Acidic gases such as sulphur dioxide and nitrogen oxides (1)
dissolve in water vapour in the air (1)
to form sulphuric acid and nitric acid, which makes rainwater more acidic. (1)

(ii) Acidic gases that form acid rain can be carried by wind to places thousand miles away from the sites of release. Therefore, it may cause acid rain at distant locations. (1)

(Total: 13 marks)

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